# Introductory Data Analysis

Some questions we ask:

1) Does misorientation angle impact average thermal resistance ($R_B$ averaged across the GB) for coincident site lattice (CSL) GBs?

We just reproduce the plot from Heat Transport at Silicon Grain Boundaries (Isotta et al. 2024).

2) How much variation in $R_B$ is present across the same grain boundary?

We might consider the minimum value. Remember that according to (Isotta et al. 2024), "We remark that each image corresponds to the average of three e-images of the same region, in order to reduce the error that stems from incoherent noise in the measurement," and of section 2.1:

3) Is the variance in $R_B$ across GBs meaningfully different based on properties of the GB?

We might consider the misorientation angle, roughness, nanostrengthening, or angle from GB to sample vertical, as in (Isotta et al. 2024). We create a new dataset linking each region (241-873) to these features. These can be read in with `CG_prep_CSL.pkg`.

To get the sample vertical, I posted the image into vector software (Figma) and tried to match the line angle. I'd say it's accurate within plus or minus 5 degrees. These are stored in `cleaned_si_data/area_properties.csv`.

Some GBs were "faint sharpest". In this case, I took the angle of the line from the beginning of the red dotted line to the end, although this may not be a good idea. The "faint sharpest" grains were A8, A11, and A3. We might consider splitting A3 in particular.

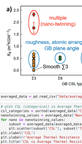
**For Dumping:**

1. Are the provided data points in the data pos provided on also an average of three values?
2. Would you be able to check over the `cleaned_si_data/area_properties.csv`? The end-make sure I correctly specified the misorientation angle, roughness, nanostrengthening, and angle from GB to sample vertical for each?
3. Do you have any other ideas for quantifying the coincident boundary (see `rae_si_data/MED_A38/Vis/GB/MED/visc/F_MB_original_pair1.png`). This can cause the estimate of the mean and variance to be affected. Based on your physical understanding, does this seem to be due to some physical defect in the property (at that point in space, there's some crystal impurity that has a very different $R_B$ value), or some sort of systematic measurement error (since this is FOTR, some momentary issue with the laser)?
4. Should we split A3 into left GB and right GB, similarly to A2?

```
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import shapiro, median_abs_deviation
```

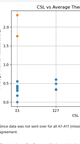
## Question 1: Reproducing Plot from Heat Transport at Silicon Grain Boundaries (2024)

As the paper mentions in the abstract, "coincident site lattice boundaries exhibit uniform excess thermal resistance along the same boundary." Therefore, we expect $R_B$ not to change very much with CSL type.




```
averaged_data = pd.read_csv("data/averaged_si_data/Polished_10kmag/si.csv = Cleaned.csv")
```

```
# plot CSL (categorical) vs Average Thermal Resistance (numerical) for each CSL category
csl_categories = sorted(averaged_data["CSL"].unique())
nanostrengthening_values = averaged_data["Nanostrengthening"].unique()
for nv in nanostrengthening_values:
    subset = averaged_data[averaged_data["Nanostrengthening"] == nv]
    plt.scatter(subset["CSL"], subset["Thermal Resistance"], label=nv)
plt.xlabel("CSL")
plt.ylabel("Average Thermal Resistance (m_K/W)")
plt.title("CSL vs Average Thermal Resistance")
plt.grid(True)
plt.legend(title="Nanostrening")
plt.show()
```




Since data was not sent over for all A1-A11 (missing A1 and A6), and there are multiple grain boundaries included for A2, A10, A8, and A9, the plots slightly differ, but this is overall in agreement.

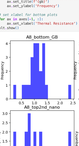
## Question 2: Quantifying Variation Across GBs

```
# plot std dev of thermal resistance for each gb site
plt.figure(figsize=(15, 6))
for gb in gb_sites:
    gb_data = averaged_data[averaged_data.index == gb]
    std_dev = gb_data["Thermal Resistance"].std()
    std_devs.append(std_dev)

# plot CSL vs std dev of thermal resistance
plt.figure(figsize=(15, 6))
for gb in gb_sites:
    gb_data = averaged_data[averaged_data.index == gb]
    plt.plot(gb, gb_data["Thermal Resistance"].std(), 'o')
plt.xlabel("GB Site")
plt.ylabel("Standard Deviation of Thermal Resistance")
plt.title("Standard Deviation of Thermal Resistance vs GB Site")
plt.grid(True)
plt.show()
```

```
# shapiro-wilk test for normality for each GB site
# H0: data is normally distributed
alpha = 0.05 # typically
print("Shapiro-Wilk Test:")
for gb in gb_sites:
    gb_data = averaged_data[averaged_data.index == gb]
    statistic, p_value = shapiro(gb_data["Thermal Resistance"])
    print(f"{gb}: Statistic={statistic:.4f}, p-value={p_value:.4f}")
    if p_value < alpha:
        print("  Data is not normally distributed")
    else:
        print("  Data is normally distributed")
```

```
Shapiro-Wilk Test:
A8_bottom_GB: Statistic=0.9600, p-value=0.7500
  Data is normally distributed
A8_bottom2nd_nano: Statistic=0.8750, p-value=0.2188
  Data is normally distributed
A8_top2nd_nano: Statistic=0.9167, p-value=0.4375
  Data is normally distributed
A10_bottom_GB: Statistic=0.8333, p-value=0.1250
  Data is normally distributed
A10_top_GB: Statistic=0.8333, p-value=0.1250
  Data is normally distributed
A2_left_GB: Statistic=0.8667, p-value=0.3125
  Data is normally distributed
A2_right_GB: Statistic=0.8667, p-value=0.3125
  Data is normally distributed
A9_bottomGB: Statistic=0.8667, p-value=0.3125
  Data is normally distributed
A9_middle_nano: Statistic=0.8667, p-value=0.3125
  Data is normally distributed
A1: Statistic=0.8667, p-value=0.3125
  Data is normally distributed
A6: Statistic=0.8667, p-value=0.3125
  Data is normally distributed
A7: Statistic=0.8667, p-value=0.3125
  Data is normally distributed
A11: Statistic=0.8667, p-value=0.3125
  Data is normally distributed
```




It seems like A10, A9, A3, and A4 have distributions that seem non-normal. This mostly seems to be caused by extreme outliers.

To deal with the outliers, we may consider using robust statistics (median instead of mean, median absolute deviation instead of standard deviation).

We now plot the MAD instead of the standard deviation. This results in significantly lower estimates of the "variation" within a single GB.

```
# plot median absolute deviation of thermal resistance for each gb site
median = []
for gb in gb_sites:
    gb_data = cleaned_data[cleaned_data.index == gb]
    # get median absolute deviation
    mad = median_absolute_deviation(gb_data["Thermal Resistance"], nan_policy='omit')
    median.append(mad)

# plot CSL vs median absolute deviation of thermal resistance
plt.figure(figsize=(15, 6))
plt.plot(gb_sites, median, 'o')
plt.xlabel("GB Site")
plt.ylabel("Median Absolute Deviation of Thermal Resistance")
plt.title("Median Absolute Deviation of Thermal Resistance vs GB Site")
plt.grid(True)
plt.show()
```

## Question 3: Properties of GBs affecting Variance of $R_B$ across it?

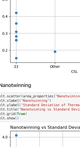
We plot CSL misorientation angle, nanostrenning, and angle to vertical vs. variance of $R_B$ across the GB.

```
area_properties = pd.read_csv("data/cleaned_si_data/area_properties.csv", index_col=0)
gb_sites = area_properties.index.unique()
std_devs = cleaned_data.groupby('GB_site')['Thermal Resistance'].std()
mean = cleaned_data.groupby('GB_site')['Thermal Resistance'].mean()
median = cleaned_data.groupby('GB_site')['Thermal Resistance'].median()
```

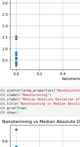
### Misorientation Angle

We can't half much since there are only three Σ27 GBs and a single Σ9 GB. **We should collect more data for these CSLs.**

```
plt.scatter(area_properties['CSL'], std_devs[gb] for gb in gb_sites)
plt.xlabel('CSL')
plt.ylabel('Standard Deviation of Thermal Resistance')
plt.title('CSL vs Standard Deviation of Thermal Resistance')
plt.grid(True)
plt.show()
```

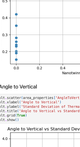



```
plt.scatter(area_properties['CSL'], [mean[gb] for gb in gb_sites])
plt.xlabel('CSL')
plt.ylabel('Median Absolute Deviation of Thermal Resistance')
plt.title('CSL vs Median Absolute Deviation of Thermal Resistance')
plt.grid(True)
plt.show()
```

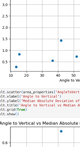



### Nanostrenning

```
plt.scatter(area_properties['Nanostrenning'], std_devs[gb] for gb in gb_sites)
plt.xlabel('Nanostrenning')
plt.ylabel('Standard Deviation of Thermal Resistance')
plt.title('Nanostrenning vs Standard Deviation of Thermal Resistance')
plt.grid(True)
plt.show()
```




```
plt.scatter(area_properties['Nanostrenning'], [mean[gb] for gb in gb_sites])
plt.xlabel('Nanostrenning')
plt.ylabel('Median Absolute Deviation of Thermal Resistance')
plt.title('Nanostrenning vs Median Absolute Deviation of Thermal Resistance')
plt.grid(True)
plt.show()
```




### Angle to Vertical

```
plt.scatter(area_properties['AngleToVertical'], std_devs[gb] for gb in gb_sites)
plt.xlabel('Angle to Vertical')
plt.ylabel('Standard Deviation of Thermal Resistance')
plt.title('Angle to Vertical vs Standard Deviation of Thermal Resistance')
plt.grid(True)
plt.show()
```

```
plt.scatter(area_properties['AngleToVertical'], [mean[gb] for gb in gb_sites])
plt.xlabel('Angle to Vertical')
plt.ylabel('Median Absolute Deviation of Thermal Resistance')
plt.title('Angle to Vertical vs Median Absolute Deviation of Thermal Resistance')
plt.grid(True)
plt.show()
```




I don't see anything particularly suggestive in these plots, although more data would definitely help!